

Sensitive Information: A Review and Research Agenda

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A basic challenge for intelligence and security informatics is exploring the ways in which humans categorize or classify “sensitive” information. Research and practice support the critical nature of categorization frameworks, yet there are a number of different ways humans conceptualize sensitive information. In this paper we review some of the dilemmas associated with classification of sensitive information, present different classification approaches, and then identify alternative propositions related to factors that influence judgments about degree of sensitivity. We conclude with directions for future research.

Introduction

A basic but critical challenge for intelligence and security informatics is exploring the ways in which humans categorize and classify sensitive information. There is a need for research that seeks not only to understand human conceptualizations of sensitive information but also to discover ways that sensitive information differs from other organizational information, and to discover what factors influence judgments about the degree of sensitivity. Our focus in this paper is to explore the literature and conclude with a synthesized direction for future research.

The literature suggests that there are a number of different ways that humans conceptualize sensitive information. Sensitive information can be regarded as an asset or resource. It can be protected by a regime of elaborate rules specifying who has the right to control the information. It can also be maintained by the social inhibitions defined by social relationships. It can be protected merely because its disclosure affects the image that others have of us or of our organization. The problem is that it is unclear if workers

make judgments about sensitive information on the basis of one, several, or all of these concepts.

This need to understand sensitivity cues is prompted by recent developments in the information technology (IT) field. Specifically, it has been prompted by some proposed security solutions that require organizations to segregate information on the basis of sensitivity (PoliVec, 2002) or to identify sensitive information in organizational communications (Liddy, 2001). Even the U.S. federal government has repeatedly voiced a need for research to understand classification of sensitive information. Since these gaps in research focus squarely on the information, research to fill them will complement parallel research efforts related to information systems security architecture, especially systems with multiple levels of security. In addition, this research will inform other content-related research streams such as filtering, archiving, tagging, and declassification.

We’ve organized this paper around four issues. First, we explore some of the dilemmas and gaps associated with the classifying of sensitive information for security purposes. In the second section we review dominant categorization and classification approaches used by humans to classify sensitive information. Then we explore five propositions about influences on the cueing or tagging used in classifying information as sensitive. In the final section we propose an agenda for future research.

Information Sensitivity: Dilemmas and Gaps

A review of the literature identified nine dilemmas and associated gaps related to classification of sensitive information. These gaps include weaknesses in current security classification schemes, as well as a lack in certain types of research associated with information sensitivity. Each dilemma is addressed in full, following the summary presented in Table 1.

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TABLE 1. Summary of information sensitivity dilemmas.

Dilemma	Description
Dilemma #1	Current government classification systems have evolved incrementally and are lacking overall coherence.
Dilemma #2	Understanding the semantic jungle needed to clarify “sensitivity.”
Dilemma #3	The technological imperative ignores limitations of embedded schemas.
Dilemma #4	Network security challenges complicate classification decisions.
Dilemma #5	Boundary-based views of information security ignore human “boundary crossers.”
Dilemma #6	Inward views of differential security require increased number of classification decisions.
Dilemma #7	Information ownership and sharing increases number of people involved in classification decisions.
Dilemma #8	Human judgment in classification of sensitive information is poorly understood and under-researched.
Dilemma #9	Sensitivity judgments and classification within LIS research are increasingly interdependent, yet little understood.

Dilemma #1: Current Government Classification Systems Are Lacking

The system of classification of national security information in the United States government has been developed over many years of use. It is a classification system that has continually changed to adapt to new requirements. Several public events and studies of note have highlighted some aspects in the organization of that information and revealed some of the difficulties in organizing this type of information. While the U.S. government is generally considered to have the most advanced security classification system, there has been a steady chorus promoting the need for fundamental research on the security classification system. Several representative instances are provided below.

In 1985 the U.S. Secretary of Defense (SecDef) established a Department of Defense (DoD) Security Review Commission in the wake of the discovery of an ongoing breach of security caused by the activities of several members of a so-called “family of spies,” (Earley, 1988) led by a former U.S. Navy sailor, John Walker. The Commission’s final report to the SecDef provided a number of recommendations. Among the areas highlighted for future research was research into how the classification actually works in practice (e.g., How much improper classification is there?), how much classified information is created, how much is destroyed, and how to provide a clearer basis than presently exists to manage the system.

A Joint Security Commission (JSC) was convened in 1993 to develop a new approach to security for information related to national security. The JSC’s report in 1994 (Joint Security Commission, 1994) recommended some sweeping changes for a classification system that had “grown out of control.” This report states that the “greatest weakness” is that “critical

specially protected information . . . is not clearly identified” (GAO, 2000, p. 14). This Commission also recognized a coordinated research and development (R&D) effort was needed to cover both the classified (national security information) and unclassified systems. It is unclear how many of these reviews were subsequently accomplished.

Recently, two reports have emphasized the need for reform. The U.S. General Accounting Office (U.S. General Accounting Office [GAO], 2000) report addressed government-wide information security policy. The report highlights the need for a set of data classifications that could be used by all federal agencies to categorize the criticality and sensitivity of the data they generate and maintain. The report offers a suggested range of categories that could be used (p. 11). A joint report by intelligence committees of both houses of the U.S. Congress investigated the intelligence related to the terrorist attack on September 11, 2001 on the World Trade Center in New York and the Pentagon in Washington, DC. These committees saw fit to recommend that Congress “review the statutes, policies, and procedures that govern the national security classification of intelligence information” (U.S. Congress, 2003, p. 14). Research into formal classification schemas used in other types of organizations is virtually nonexistent.

Dilemma #2: Unscrambling the Semantic Jumble

The dilemma of semantics and information security classification has also become an issue. Following the terrorist attacks on the World Trade Center (WTC) and Pentagon in 2001, the U.S. federal government took action to eliminate all information on government-affiliated information sources that could aid terrorist attack planning (Study: Most Support, 2002). This included information posted on government Web sites and stored in repository libraries across the U.S. These actions highlighted a recurring difficulty (Dilworth, 1987)¹ in the classification of information in the federal government; namely, a category of information that is sensitive but does not come under the definition of classified information under national security directives. This type of information is sometimes called “sensitive but unclassified.” The attacks on the World Trade Center and Pentagon served to bring this issue to the fore once again. In addition, it cast in a new light other information reposing in government databases not previously thought to be sensitive or critical to national security. Few could have guessed the importance of information collected and stored as a part of the daily operations of the Immigration and Naturalization Service. More recently, the issue has emerged with respect to restrictions on the dissemination of the results of government-funded academic research that could aid terrorists (Congressional Research Service, 2003).

¹Dilworth’s article was part of a continuing debate in the *Bulletin of the American Society for Information Science* [13(4), 12–16; 13(5), 10–11] relating to an earlier attempt by the U.S. Government to create this new category.

Dilemma #3: The Technology Imperative

Scholz (1990) suggests that the values of system developers and/or champions are embedded in the design features of new systems. This can be inferred to include security systems and network controls, especially those with a broad scope like those found in public key infrastructure (PKI).² As such, cues as to sensitive information or access are not only embedded in explicit security measures, but also in the implicit technical constraints. Examples of this would include rules about who has access to features of different systems, the nature of data that is available to employees at different levels, and user interface design that makes it easier or more difficult to locate and access stored information. These embedded schemas can often collide with existing schemas in the workplace. Organizations and their members try to incorporate these embedded guidelines as to what is sensitive or secure and may abdicate personal decision making to the technological imperative. Unfortunately, existing guidelines for classifying information as sensitive are more complex than can be easily incorporated into systems. Similarly, much of the information that may be "sensitive" is not guarded by technology, or embedded controls can be easily overruled by technical necessity at many points within the system.

Dilemma #4: Networked Security Challenges

New threats like destructive software, hackers, and terrorist attacks have added new dimensions to the task of network security. In like manner, threats that are more traditional have been made easier with the new technology, such as insider theft and fraud. All of this has prompted organizations to be less confident in their electronic networks and the security of the information coursing through them. It has also introduced a whole new category of sensitive information, namely, all the information surrounding network security controls.

There is ample evidence of the major difficulties for workers dealing with network security in organizations. Surveys have documented these difficulties and have purportedly revealed that the problem does not appear to be getting any better.³ Ryan's (2001) investigation of small and medium-sized businesses showed that the information technology itself adds to the difficulties because of its complexity and the need for constant attention. To make matters worse, the difficulties in network security have occasioned a veritable deluge of additional new technology to help network security workers.

²Public key infrastructure is a set of processes and procedures to manage encryption keys and validation documentation in an organizational setting.

³Surveys are conducted jointly by the Federal Bureau of Investigation of the United States in conjunction with the Computer Security Institute (CSI) and PriceWaterhouseCoopers Corporation. The publishers of *Information Security* magazine have conducted surveys annually since 1998.

Dilemma #5: Boundary-Based Information Security Models

Coping with security challenges has become a major preoccupation within the information technology (IT) industry. While many are persuaded that security problems will eventually be solved by more technology, others have called into question the basic models of security being employed in today's networks (Jones, 2002). Specifically, models based on the systems concept of making boundaries airtight may no longer be viable. This is primarily because humans, a critical and sometimes forgotten element of the overall information system, have a habit of moving back and forth across the various boundaries. As an example, humans traverse the physical boundaries with small portable computers carrying organizational information with them. They transcend electronic boundaries through their searching behaviors, bringing computer viruses and destructive software in from the outside. Drake and Clarke (2001) explore these issues from the standpoint of social theory and suggest that network security is largely a human problem.

This aspect of network "insecurity" was highlighted in 1997 when the United States Department of Defense (DoD) conducted a worldwide exercise to test network defenses. This exercise, known by its DoD code name, *Eligible Receiver*, demonstrated that network defenses, although designed to be hard on the outside, could not detect or prevent extensive penetration after the initial shell was broken. This has led analysts to use a metaphorical phrase "crunchy on the outside and chewy in the middle" to describe current network architecture that offers some resistance at first, but is easily penetrated and circumnavigated once cracked.

Defining security and insecurity leads to the next dilemma, which is differential security.

Dilemma #6: Inward Views of Differential Security

Several models of network security have been developed which focus inward on the network and its functions to make it more secure. Two aspects of these "inward-looking" models are continuous monitoring of the network (Hare, 2001, p. 568) and reconfiguration of the organizational network (Murray, 2000, p. 223) so that *differential security* can be provided to the different parts of the network. Various ways have been suggested to accomplish these tasks. Some are focused on the network hardware and software components. Another is to focus on the information itself. As a preliminary step in this focus, security experts have suggested that organizations need to determine the protection needs of the information, that is, they need to determine which information requires the most protection and which the least protection based on the sensitivity of the information (British Standards Institute, 2000; Eloff, Holbein, & Teufel, 1996). These authors and others (Parker, 1996; Peltier, 1998) have also emphasized the importance of a classification system for information to realize this goal.

While members of the IT field have suggested this need to perform a sensitivity assessment, other recent events lead

us to believe that organizations may already have a rough idea of the differential protection needs for information in both electronic and manual systems (Farrell, 2002). In recent cases involving a number of organizations such as the Enron Corporation (Iwata, 2002); Arthur Anderson (Farrell, 2002); Merrill-Lynch (Valdamanis, 2002); ImClone (O'Donnell, 2002); Firestone Tire and Rubber Company (Nathan, 2000); and the venerable Roman Catholic Church (Sheler & Glasser, 2002), organizations appear to have a good feel for the ways that differential protection of information might be accomplished. Journalistic accounts suggest that individuals in the organizations knew, only too well, what information was sensitive and what was not. In one case, that of Arthur Anderson, this information became an overriding concern of a number of key individuals trying to cope with the scandal. In this case, individuals performed wholesale destruction of incriminating documents before investigators could get to them.

These recent cases highlight the fact that judgments about sensitive information are not as simple as one might think. Paradoxically, each of these cases could be used to argue that information security was too lax or that information security was too tight. To those inside these organizations, the security was too lax because damning and harmful information was made known outside the organization. To those outside these organizations, the security was too tight because the information revealing corporate wrongdoing was bottled up inside, creating a situation that was harmful to employees and, in some cases, to the public as well.

These two views are not mutually exclusive. It is also not correct to suggest that organizations do not know what is most sensitive and what is not. On the other hand, many organizations have not seen the need, at the organizational level, to specify and organize the information based on sensitivity criteria, as have other organizations such as some agencies of the federal government. One survey of 4500 security professionals in 42 countries found it "surprising" that 39% of the organizations participating in the survey did not classify their sensitive information (Hulme, 2001).

Dilemma #7: Information Ownership

The post-mortem analysis of failings leading up to the attacks of September 11th focused attention on another recurring problem in our networked world, and that is the inability or the unwillingness to share information with other groups or organizations. The impetus for the creation of the new Homeland Security Department was, in part, recognition of the need to cut through the bureaucratic logjam that would not allow interdepartmental and interorganizational information sharing. While information sharing is generally considered to be a desired goal, it contravenes a timeless security principle, namely, the greater the access to the information the less secure it becomes. There are also countervailing considerations that cause equal or greater concern, such as the need to maintain the privacy of individuals.

Dilemma #8: Human Judgment in Classification

Although most of the literature presumes rationality in decision making, research into interpretations of information technology has indicated that the same systems can be interpreted differently (Brown, 1995; Kaarst-Brown & Robey, 1999; Orlikowski, 1991). Similar interpretations, referred to as sensitivity judgments, have been suggested as critical in proposed systems used to classify electronic documents (Eloff et al., 1996). Such classifications are: "... Highly dependent on human judgment, because it requires someone to interpret the content of the document" and that the content "... must be interpreted and categorized according to its sensitivity" (Eloff et al., 1996, p. 58).

Although, Eloff et al. consider this categorization to be a key aspect of a classification system, they do not explore further the meaning of sensitivity or how one might go about constructing these categories or, more importantly, discovering the categories already in use. While the phenomenon of sensitive information appears to be ubiquitous, a number of its aspects are unclear. It is uncertain if sensitive information refers to (a) a separate type or category of information, (b) a sensitivity encoding (tag, if you will) that humans apply to all information, or (c) contextual information about sensitive situations and/or sensitive subjects.

This distinction is critical. If, for example, humans infer sensitivity by merely detecting certain keywords in the text, rules can be constructed and algorithms developed that will guide future classification assignments for similar information. If sensitivity is derived from a complex set of rules, both tacit and explicit, then those rules can be used to build a formal classification system. If, on the other hand, sensitivity cannot be inferred from the text or derived from a set of rules but must also take into account the surrounding context, sensitivity determinations are considerably more complex and require an understanding of which cues individuals are more sensitive to.

Dilemma #9: Sensitivity Judgments, Classification, and Library and Information Science Research

This real-world problem regarding sensitive information relates directly to a more abstract problem in the library and information science (LIS) world. Library and information science researchers investigate tagging and categorizing as central issues in the organization of information. This research has traditionally supported the work of those who perform classifications (indexing, abstracting, cataloging, and so forth), and it has also delved into the classification practices of everyday workers (Kwasnik, 1991).

In the LIS world, these tags (labels) reveal some attribute of the information or the content. An example is the functionality programmed into most electronic mail systems that gives the user the ability to specify an attribute of the e-mail that is not readily apparent from the text of the message itself. Two such functions are (a) the priority function and (b) the function that indicates if a message might contain

some objectionable content. In the case of this sensitivity attribute, this tagging or labeling does not describe the subject (*aboutness*) of the content but instead describes some other quality of the information.

Synthesizing the Dilemmas Around Information Sensitivity

We have synthesized the diverse dilemmas identified above around the concept of “information sensitivity.” One research question that is central to the above-noted dilemmas is how humans organize information on the basis of sensitivity. A second research question is related to whether classification decisions based on different factors result in assessments that some sensitive information is *more* sensitive than other sensitive information. The following section reviews some dominant approaches to human classification and categorization of information, and how it can inform future research into information sensitivity.

Classification and Organization of Sensitive Information

Theory and research related to how humans organize information is central to future research agendas on intelligence and security informatics. Library and information science research deals specifically with classification systems that humans invent and maintain to be able to find information they need. This is of particular relevance to our understanding of information sensitivity decisions. Five aspects of classification will be reviewed. The first aspect involves formal systems for organizing, otherwise referred to as “classification schemas.” The second aspect is the development of categories of information. The third aspect explores concepts of forgetting, monsters, borderlands, and the “other” category. The fourth aspect reviews literature on the relevance of subject, and addresses concepts of “aboutness” and differential relevance found in texts, intertext, and context. The final aspect included here is about rules, schemes, and existing frame knowledge.

Formal Classification Systems

Kwasnik (1999) outlined various schemas by which information or knowledge can be organized into formal systems. These systems are hierarchies, trees, paradigms, and facets. She points out the principles of organization underlying each and the advantages of each for organizing knowledge. Her work is not focused on organizations per se, but it is relevant for future research agendas. Table 2 lists the types of classification systems and selected characteristics. It provides an excellent summary of the ways that knowledge is organized. Kwasnik makes the point that these systems can also be useful in knowledge discovery as well. Beghtol (1995) reinforces this point.

These classification schemas are representative of possible ways to think about organizing sensitive information.

TABLE 2. Classification systems.

Classification scheme	Primary relationship	Principles
Hierarchy	Form an “is-a” relationship from top to bottom	Inclusiveness, species/differentia, inheritance, transitivity, systematic rules for association, mutual exclusivity, necessary and sufficient criteria.
Tree	Systematic relationship but no “is-a” relationship	Complete and comprehensive relationship, systematic and predictable for distinction, citation order.
Paradigm	Entities described by two attributes at a time	Two-way hierarchical relationship, axes represent two attributes of interest, cells may be empty or have more than one entity.
Facet	No primary relationship—Recognition that complex entities have many facets	Does not require complete knowledge, hospitable, flexible, expressive, does not require strong theory, can accommodate a variety of theoretical structures and models, multiple perspectives.

They are, accordingly, schemas that are encountered in the workplace and could emerge in future research. There is one important lesson from examining these classification schemas. Not all schemas are neat and clean with a fully developed hierarchical structure. This lack does not mean that there are no organizing principles at work, however.

Others have hinted that the organization of information in organizations consists of a complex mixture of formal and informal systems that co-exist within the organization (Bowker & Star, 1999). A number of such systems force a formal structure, such as those that arise from rule-bearing dicta. Others arise as the members of the organization attempt to make sense of their environment and to communicate with each other about that environment.

Development of Categories

Categorization and classification are closely intertwined. Jacob (1991) makes a cogent argument for the distinction between the two. It is apparent that, at best, we can say that once a category is formed in the mind, a classification schema is born at the same time. It is impossible to say which comes first—classification or categorization. The very process of formulating the category entails a classificatory thought process deciding what is “in” and what is “out.”

Thinking about human categorization has changed greatly in the past 50 years. Prior to the 1950s, categories were thought to be rigid and immutable. Based on Aristotelian concepts of organizing the world, every member of a category was distinguished by one or more well-defined descriptors or characteristics (Taylor, 1999). This view has been questioned, however. Wittgenstein (1953)

exposed apparent weaknesses in the Aristotelian view of categories by suggesting that they were not well defined and clear but were instead only approximate indicators of how humans think about their world. He introduced the notion of family resemblance to convey his assertion that category members are not all the same but have certain characteristics that suggest they belong to the category. This way of looking at categories also implies that humans will have a high degree of agreement about the members of a category, but that the agreement will not be unanimous. Unfortunately, it also hides the fact that we can categorize a set of humans in many different ways. Members of a family can be shown to have certain resemblances, but they can also be shown to have other characteristics (personality, for instance) that make sense in differentiating between them (Brown, 1958).

Recent categorical research, conducted in a number of different fields, has provided some interesting insights, as well as new techniques, for examining the phenomenon of information sensitivity decisions. The primary insights for the purposes of future research are summarized in Table 3.

TABLE 3. Notable aspects of human categorization.

	Description
Fundamental activity	It is uncertain if this occurs because of the way that the brain functions or arises from the formation of languages (Markman, 1989).
Cultural differences	There are cultural differences in the formation of categories. For example, cultures apparently vary in the number of colors that they consider base colors, from 2 to as many as 12 (Berlin & Kay, 1969).
Differences in category definition	Some categories are well defined and some are not (Zadeh, 1965).
Cognitive economy	The formation of categories affords a cognitive economy in ordering the experiences and objects of everyday life (Rosch & Lloyd, 1978).
Base level	Within a category, there is a base level which humans use for ease of communication about the category. For example, in biology the classification level of genus is considered the base level for talking about the classification of plants or animals (Brown, 1958; Mervis & Rosch, 1981).
Membership	Humans vary in their agreement about membership in categories. For example, an ostrich may not always be considered a member of the category of birds. This is called graded structure or graded typicality effects (McCloskey & Glucksburg, 1978).
Best example	Humans often agree when asked to choose the best example of a category when given the name of the category and several choices from that category (high cue validity). An example is that some shades of a color are more often picked when humans are given a set of members from the category and asked to pick an example of that color (Rosch, 1978).

Forgetting, Monsters, Borderlands, and the "Other"

There are four other aspects of the broader use of categories in organizations that have arisen in previous research and that are relevant to future research agendas. First is the concept of "forgetting" (Bowker & Star, 1999), whereby previous practices or classification decisions are discarded and "forgotten" as new practices and new rules are put in place. An example is the sudden change in the protection of information related to a merger or acquisition in the private sector. The second and third concepts are related: "monsters and borderlands." A monster occurs when an object refuses to be "naturalized"⁴ (Haraway, 1992). We interpret a "monster" as sensitive information that exists in two or more different classification schemas, or in the schemas in two different organizations, and which has a slightly different meaning in each. An example is information about a hacker attack on a corporate network. Corporations are loath to share this information without some guarantee that it will not find its way into the public domain. A borderland occurs when two communities of practice co-exist in one person (Anzaldúa, 1987). A borderland can be conceptualized as an individual who understands two or more different classification schemas or one who knows two or more different meanings for a monster. An example would be an intelligence analyst who is "plugged in" to both the law enforcement and the intelligence communities. Finally, the concept of the "other" category is important. The "other" category is the group or collection of things that do not "fit" into a classification schema. There is value in studying the "other" for what it reveals about the classification system as a whole. An example is information that is marked "for official use only" in both the public and private sectors.

Aboutness and Differential Relevance: Text, Intertext, and Context

The classification/categorization literature informs the process by which categorization judgments are made. Beghtol (1986) suggests that there are different types of document aboutness. She highlights the works of Fairthorne (1969), Boyce (1982), and Van Dijk (1979), all of whom use different terminology to get at distinctions between aboutness and meaning. Beghtol goes on to look in depth at Van Dijk's theory of aboutness. Included are two concepts that relate directly to the classification for security purposes outlined by Eloff et al. (1996). The first is the concept of intertextuality, whereby the textuality of any one text arises from the interaction with other texts. The second is the concept of extratextual knowledge (a.k.a. frame knowledge) that is "conventional knowledge, brought from the readers mind to bear on the text." Beghtol asserts that there is "one special type of frame knowledge that readers bring to texts." That is their "expectations for both the content and the structure of

⁴Bowker and Star define naturalization as stripping away the contingencies of an object's creation and its situated nature.

different kinds of texts (Beghtol, 1986, p. 89).” This appears to align very closely with Eloff and colleagues’ concentration on both content and structure (Beghtol, 1986, pp. 84–88).

Of particular interest to the issue of the meaning of sensitivity is Van Dijk’s (1979) suggestion of a “differential relevance” assignment after the normal “aboutness” of the document has been judged. The differential relevance assignment occurs when one differentially selects items for “special treatment from among similar items” if it results from a greater interest, an immediate purpose, or a special task. Van Dijk asserts that *special purpose overrides* text structure during comprehension. It would seem reasonable to suggest that a special task could be conceived to be the assignment of a sensitivity metric (Beghtol, 1986).

The above literature suggests that meaning with respect to the classification of information for security purposes can be seen to have elements of the textual, intertextual, and extratextual (frame knowledge) (Beghtol, 1986). The question is whether this assignment arises from one, two, or all three of the above.

Rules, Schemes, and Existing Frame Knowledge

Research in organizations provides an indication of the frame knowledge brought to bear on the categorization decision-making. Beghtol (1986) suggests relevant frame knowledge consists of rules, schemas, etc. Research by Cyert and March (1963) provided the conceptual justification for the examination of rules to reveal the elements related to the meaning of sensitive information in organizations. We interpret these rules as part of intertextual cues that individuals incorporate into their sensitivity decisions.

Liddy indicates that an examination of the business rules can provide a basis for categorizing the information with respect to sensitivity (Liddy, 2001). This approach in the information science world seems to connect in the organizational science world with the notion of rules in organizations. March, Schulz, and Zhou (2000) assert the centrality of rules in the organization as, *inter alia*, a mechanism for rational efforts to organize and as a vehicle for the construction of meaning. Future research on information sensitivity should therefore consider both written and unwritten rules. March et al. (2000) also acknowledge the importance of rules imposed on the organization by rule-making institutions external to the organization; however, they consciously exclude them from their study. As noted earlier, the existence of rules or guiding schema is still filtered through individual judgments. Therefore, security informatics research would benefit from a better understanding of the expected impact of these rules on the classification decisions of individuals in organizations.

Frame Knowledge in Classification Schemas

Four domains of research inform our understanding of the nature of sensitive information. Each incorporates a different conceptualization of sensitive information: economic,

legal, social, and psychological. Each of these domains looks at sensitive information differently and therefore provides a valid basis upon which differential protection of the information could be reasonably argued. These domains are presented as guiding propositions for future research.

Frame Knowledge—An Economic Perspective

Guiding proposition 1. Judgments about the sensitivity of information are based upon the value of the information.

Economic research has yielded understanding of an economic-based view of information. Research on the economics of imperfect information in general (Philips, 1989), and asymmetric information in particular, has dealt specifically with the thought processes underlying a differential protection schema related to the economic value of information. In this view, sensitive information is information which is protected to increase the probability of a favorable outcome for the person, group, or organization that controls that information, or to preserve or increase the options for future action or decisions (Gennotte & Trueman, 1996). In this case, information is an asset, resource, or, in some cases, a commodity that can be used, bought, acquired, and sold (Bushman, 1991). It can also be conceptualized as a signal (Pae, 1999), normally a signal of market conditions or status of one element of the market. It becomes a signal when it is made known to others. Otherwise, it remains protected and known only to a restricted audience. Its value derives from the advantage that accrues from having information that others do not have (Aboody & Lev, 2000).

Information asymmetries are of particular interest because they occur naturally and because they contribute to the dynamics of the marketplace. The absence of such asymmetries contributes to market transparency (Bloomfield & O’Hara, 1999). This domain of research suggests that information asymmetries and sensitive information may be the same phenomenon or conceptual cousins. Much of the research in the economic domain is done at the level of the firm, and focuses on the interface (boundary) between the firm and the outside world, as well as the information flows across that boundary.

Information flows are generally couched as issues of disclosure for the firm (Teoh, 1997). This presupposes its antithesis, *enclosure*, which could be the same as secure information. The firm is generally proposed to have control over the flow and makes conscious decisions about the movement of information from the inside to the outside based on the implications (usually gain/loss) of that flow into the marketplace. These flows are not always made voluntarily. Teoh (1997) talks about the impact of nondisclosures and adverse (forced) disclosures as signals of the firm’s value.

Temporal change (Kim & Verrecchia, 1997) and the timing of disclosures (Chen & Mohan, 1994) are central issues in economic research. Not only does sensitivity change with time in some human activity, it also changes as soon as the disclosures are made known in the public arena. Using value

as the criterion for classifying information for security purposes would require the establishment of an assessment regime to adjudicate the relative value of information to the organization. This would force a continual focus on the value of information. Information would have to be judged on a relative scale in comparison with other information.

Frame Knowledge—A Legal Perspective

Guiding proposition 2. Judgments about the sensitivity of information are based upon who has the right to use the information.

Research into the institutional/legal arena has provided understanding of a view of sensitive information that is based on a system of “rights.” This conceptualization grows out of post-enlightenment political philosophy incorporating Rousseau’s ideas of a social contract and John Locke’s (1689) ideas about natural rights, in general, and property rights in particular. This philosophy is reflected in the philosophical underpinning for the U.S. Constitution, and many of its tenets are incorporated in the blueprint for governing in the US. Some of this research dovetails closely with economic research, especially when property rights are being examined. Differential protection of information is maintained by a complex set of rules whereby someone with ownership rights of the information has the right to retain it or to divulge it as that owner sees fit.

In this domain, sensitive information is simply information that the owner (the entity that has the right to the information) does not want to reveal to others. This perspective of sensitive information is very strong, of course, in the case of information related to privacy.

Two prominent classification systems are of note. One is the classification system associated with the laws and regulations in the legal system. The other is the classification system employed to protect information related to national security (discussed in Dilemmas #1 and #2 above) or to protect intelligence assets.

With regard to the first system, legal professionals and specialists in organizing legal information have a taxonomy for organizing according to the major categories into which laws are divided; for example, property law, criminal law, torts, etc. The system of classification associated with intellectual property is another example. In this system, intellectual property is a major category with patents, trade secrets, copyright, and trademarks as subcategories.

The other primary classification system is that used in high security environments in virtually all governments of the world. The familiar system of “classified information” is an example. This system is popularly seen as having levels (confidential, secret, and top secret), but is really a complex system having many facets (levels, compartments, special access programs, etc.). This type of system is not confined to national governments, however. It is also employed at the state and local level as well. There is an inherent problem with this type of information in democratic institutions

where, theoretically, the public has a right to know what goes on inside their governmental institutions and other institutions that are supposedly self-governing. One role of government is to adjudicate the flow of information for the public good. This includes information about economic activity and public health as a domestic component of national security.

It has long been a recognized problem that these institutions are notoriously reluctant to allow any information to be made public. The government has a role in deciding if organizational disclosures should be made (Chen, 1994; Konar & Cohen, 1997; Maingot, 1995). Various aspects of this problem have been explored with respect to the impact that this secrecy has on democratic institutions in general (Thompson, 1999) and on national security (Doyle, 1999). In like manner, this problem has been explored with respect to the impact that governmental secrecy has on organizations (Lewis & Poitevin, 1997).

These perspectives have little to say about how individuals perceive the problem. Shils (1956) posited the interrelationships between secrecy, publicity, and privacy. His formulation suggests a difference between private information and secrets, both of which are protected from view of others (or is kept from public view). Warren and Laslett (1977) also probed the difference between privacy and secrecy, suggesting the categories public-secret and private-secret to distinguish between (a) secrecy in relation to the institution of politics and (b) secrecy about one’s personal life. This is an interesting hypothesis, but the question is whether or not individuals make this distinction. If they do, their personal accounts regarding their own conceptualization of sensitive information would tend to reveal that individuals make such a distinction or that they think both are subcategories of sensitive information.

Notions of temporal change and timing are also present in this conceptualization. Rights to the information do appear to change over time. Two examples are the declassification program in the U.S. federal government for classified information and the statutory requirements imposed by the Securities and Exchange Commission on corporations for the periodic (e.g., quarterly earnings) and episodic (e.g., corporate officer’s sale of stock) disclosure of information.

Frame Knowledge—A Social Perspective

Guiding proposition 3. Judgments about the sensitivity of information are based upon which social group or collective controls that information.

The social/cultural research field has a long, wide stream of research into the nature of secrets and secrecy in social groupings. The concept of secret appears to be synonymous with sensitive information, but this synonymy has not been firmly established, as suggested in Dilemma #2 above.

In this domain of literature, information is seen as a sign of membership in a group or organization, or as a shared tie in a relationship. Sensitive information has a complex

pedigree in this perspective. It can be conceptualized as a form of glue that binds relationships or memberships. Sensitive information can also be seen as a token that demonstrates membership. The keeping of the secrets is often a *test* of membership. Family secrets are a prime example of sensitive information in this perspective.

This stream of research draws from Georg Simmel's writings (Simmel, 1912) on the role of secrets and secrecy in society. Simmel suggested that society had a rather negative view of secrecy but that secrecy served many functions in society, both good and bad.

Secrets and truth telling among larger groups or collectives have highlighted the role of information in the maintenance of social bonds. Palmer (1990) studied radio communications among Maine lobstermen. Fine and Holyfield (1996) reported on a voluntary group of mushroom hunters. Both reported similar findings on the sharing of information in the community or group. Palmer highlighted the role of community status in information sharing. Of particular importance in both studies was that the volume and the specificity of the information shared was inversely proportional to the anglers' or mushroom hunters' success. Fine and Holyfield (1996) reported that the degree of sharing in the voluntary organization varied according to relative status of the members; that is, the more experienced shared only enough with the newcomers to keep them out of trouble. In both studies, members were stingy with information about potentially favorable places to search or fish. As noted in Dilemmas #6 and #7 above, sharing is a norm that is mediated by other social factors.

This concern with the security of information within and among various groups in an organization was noted by Baba (1999) in a study of software implementation in American corporations. Her study highlighted the role of trust in information sharing.

The role of secrecy in social groups is apparently learned at an early age. Katriel (1991) observed secret sharing among Israeli children. Merten (1999) suggested that there is a process of socialization into secrets in adolescent and pre-teen groups. If secrets are synonymous with sensitive information, these studies suggest that humans learn the basics of information sensitivity at an early age through their early social interactions.

Included within this research stream is work on human organizations that are entirely secret in nature. Studies have been conducted of secret societies in Western Africa (Bellman, 1984), the Taos Pueblo group of Native Americans in the United States (Brandt, 1980), Voodoo societies in Haiti (Laguerre, 1980), and the Ku Klux Klan organization in the United States (Schaefer, 1980). In these secret societies, information (and its protection and survival) is a key issue in the social relationship. Trust is a prerequisite to information access. The maintenance of group secrets is the highest form of allegiance.

The secret societies that Bellman (1984) studied serve as examples of so-called cultures of secrecy. These societies operate with a shared set of behaviors as well as a shared

view of reality that forms the core of the organization. Part of this is the manner in which information is treated or shared and, possibly, a shared understanding of what is and is not shared with others outside the organization. In other words, a set of tacit rules governs this sharing. Bellman reports a somewhat paradoxical finding in his study. He observed that the information, in and of itself, is not secret. The secret is the role that the information plays in the society and society's rituals. For example, the formulation of a supposed healing potion can be divulged, or even sold, but the revelation of the role of that potion in the society's rituals is taboo. This suggests that the protection of information about the rituals and the inner workings of an organization may be universally protected from outside scrutiny. This underscores Dilemmas #6 and #7 above.

Sensitive information in the social/cultural domain includes information that an individual has acquired about a social organization or from members of that social group which the individual feels must not be known outside the social organization. This is primarily because of its role in signifying membership and support of the organization or group. In short, the information serves as social lubricant to maintain the cohesiveness of the unit and to signal identity in the group. Obtaining access to the secrets is a sign of trust in the individual member. The keeping of the group's secrets serves as a test of affinity. The betrayal of the secrets constitutes a major punishable offense not to exclude immediate expulsion and termination of membership. In a family, sensitive information is conceptualized as information that is not known outside the family unit. This can sometimes be characterized by the term *family secret*.

The consequence of disclosure of the information is to disrupt relationships, primarily within the group or organization or to expose the organization to closer, unwelcome scrutiny. This is sometimes characterized as a betrayal of trust. Especially in family groups, the consequences can be embarrassment or stigma that will affect others in the group.

Temporal change and timing are not central issues in much of the research in this social domain; paradoxically, however, it is implicit as an issue of disclosure for so-called family secrets.

Frame Knowledge—A Psychological Perspective

Guiding proposition 4. Judgments about the sensitivity of information are based upon the interpretation of the information.

In the literature with a psychological/linguistic slant, sensitive information consists of a particular view of reality that someone either holds back entirely or selectively disseminates to others. Such revelations depend upon the need to divulge the information or on the impression (image) that the representer wishes to create or foster in the minds of the receiver. Representers appear to be intent on fostering a view of themselves that is socially normal or that meshes with their own view of themselves.

The selective dissemination of information is the key to maintaining this external worldview. Goffman (1959) provides a rather elaborate differentiation for the world of secrets. He suggests six different types of secrets: (a) dark, (b) strategic, (c) entrusted, (d) free, (e) latent, and (f) inside. It is unclear if these categories arise from his personal experience or from empirical evidence. It appears somewhat doubtful, but possible, that other individuals would make such differentiations.

As with the social/cultural perspective, trust plays a big role in the keeping of these secrets. More of the truth is shared with trusted others, especially those with whom the representer shares an intimate bond (Yovetich & Drigotas, 1999). Information is shared when there is a mutual understanding (sometimes unspoken) that the information will be closely guarded. Thus, Dilemma #5 above, related to boundary crossing, may be evident at many different boundaries at the organizational, group, and individual level.

The desire for personal privacy emerges in this literature. In contrast to the legal literature, this does not emerge as an issue of rights but, instead, as an issue of privacy at its most elemental core—the wish or choice to keep certain types of information from others. Judging from the literature, the types of information are varied, ranging from mundane consumer consumption information (Moon, 2000), to information about genetic characteristics (Gostin, 1995). In between, is deeply personal information such as information about adoption (Rosenberg & Groze, 1997) and donor insemination (Landau, 1998).

This research stream suggests a central role for information, especially information dissemination (or nondissemination), in maintaining an image of self that an individual wants the world to see. Extended to the organization, the primary conceptualization is information that affects, or is contradictory to, the organizational image that is portrayed to the world outside the organization or even between groups within an organization.

Temporal change and timing do not appear to play a large role in this conceptualization except in the implicit notion that interpretations change over time.

Frame Knowledge—An Holistic Perspective

Guiding proposition 5. Individuals often must incorporate competing cues from several knowledge frames, complicating the classification of sensitive information.

There are a number of commonalities in the above domains and research streams that can be separated out and highlighted. Table 4 shows some of these characteristics: synonymy with sensitive information, boundaries, and a concern for consequences related to having the information cross the boundary or become unprotected.

One of the most interesting aspects that appear to be common to all these views is the anticipation of how others might use or interpret the information. It is this anticipation or projection of a possible interpretation that appears to drive behaviors that cause a person, a group, or an organiza-

TABLE 4. Common features of conceptualizations.

Knowledge areas	Synonyms for sensitive information	Boundaries	Consequences of boundary crossing
Economic	Asymmetric information Private information Proprietary information Corporate secrets	Organization	Gain/loss of assets Increased costs Loss of competitive advantage Narrows options
Legal	Intellectual property Trade secrets Proprietary information Classified information	Organization	Litigation Loss of advantage
Social	Secrets Confidential information	Personal Group organization	Loss of face Embarrassment Loss of trust Loss of membership
Psychological	Secrets	Personal	Embarrassment Stigma Life change Loss of image

tion to try to keep information enclosed within the personal or common boundary.

Three other similarities are worthy of note. The first is that the concept of asymmetric information, or at least the thought process, is very similar to the thought process inherent in the other knowledge areas. The economic approach relates directly to the organizational information. On the other hand, the “presentation of self” brought out in the psychological literature seems to be directly transferable to the organizational presentation of self. Both of these approaches deal directly with the issue of disclosure/enclosure and the decision to disclose or not.

Second, both the economic approach and the institutional/legal seem to explore the issue of property rights as it relates to information. In the economic approach, the information is an asset that provides leverage in decision-making. In the institutional/legal approach, the information is viewed within the lens of ownership rights, who has it and who does not. Both of these approaches struggle with the special property of information that allows it to be reproduced and be used without being consumed. This leads to the familiar and understandable behavior that causes humans to hoard information.

Finally, there are notable similarities between the social/cultural and the psychological/linguistic traditions. One deals with one’s own secrets and the other deals with secrets held in common in a group. The natural bridge between the two lies in the keeping of secrets in dyadic relationships. It would be tempting to say that the distinction made here is only arbitrary, but it is not clear if an individual views his/her own secrets the same as he/she does the secrets of the group.

Figure 1 shows a conceptual view of these four “knowledge” domains. The Venn diagram display was chosen to

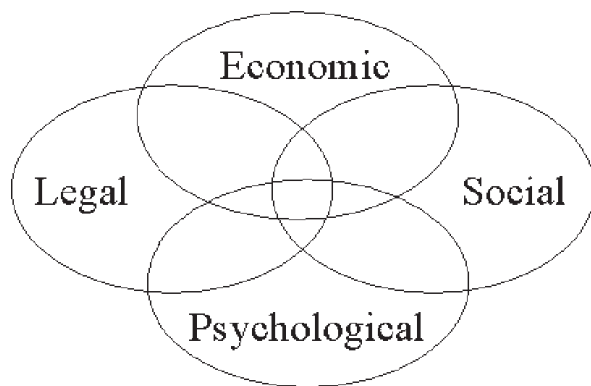


FIG. 1. Knowledge domains containing conceptualizations of sensitive information.

show the overlap between the fields and to dispel the impression that the authors' believe that these domains are anything other than interwoven.

These similarities highlight, however, that individuals may not be clear which frame informs a classification decision. In addition, the differences stand out as potentially competing frames that may impact a classification decision about information sensitivity in a way that is neither consistent across situations nor documents of the same type.

A Proposed Research Agenda

Conceptualizations of the phenomenon of sensitive information are varied and somewhat confused. No one field of research contains a conceptualization that completely describes or explains these phenomena. There is a need to attempt a synthesis of all the particular approaches. At the very least, future research requires a consideration of the various views presented here to give a full description of sensitive information. The two primary contributions presented here are that we call attention to understanding the encoding (tagging or labeling) processes with respect to sensitive information and to individuals' classification of information within the context of an organizational setting.

There is also a need to get at the individuals' understanding of this phenomenon—to understand it from his or her point of view in the way that he or she understands it. Research has shown that our preliminary assumptions about types of classifications in the workplace are faulty and ill conceived. Our understanding can be improved by providing conceptualizations that are grounded in data captured in the workplace.

To say that we need to understand how workers in organizations make sense of or interpret sensitive information is to propose that research captures thinking at an individual level; however, given contextual considerations, research needs to be done at a number of different levels to fully understand this phenomenon. Research into the ways that different classification schemas either facilitate or impede information sharing either within or between organizations would be especially illuminating.

A preliminary inquiry into judgments about sensitivity supports the use of research using qualitative methods. This approach is prompted by the urgings of other researchers in the library and information science field (Forsythe, 1992; Star, 1998). The advantage of this approach is that the categories of sensitive information are gained from humans in the workplace. As such, it is constitutive of the daily life experiences of the workers and is representative of their view. Forsythe (1992) found this type of understanding could challenge existing assumptions on which information systems development is based. Qualitative research has also been shown to be useful to understand the everyday meaning of objects and processes by observing the way in which workers talk about it and the words they use when they do. Future research using qualitative and interpretive methods would help us answer whether the words "sensitive," "secret," and "classified information" are delimiters for the same phenomenon, different phenomenon, or are sometimes the same and sometimes different depending on the situation. This gets even more interesting when one adds the concept of "privacy" and its relationships to these concepts. As presented earlier, conceptual distinctions become even cloudier when one adds competing research domain propositions such as those developed for intellectual property and other legal purposes.

Properly understood at an individual level, and grounded in individual accounts of everyday work life, it should be possible to understand an individual's sense of the sensitive. This should emerge in a coherent manner in spoken language and in the artifacts of everyday life. We can see if the accumulation of evidence based on these individual accounts yields clues as to factors that make up the sensitive, and to see if there are clues as to the whether or not the workers believe some information is more or less sensitive than others.

Since this research agenda is geared to focus on the understanding of classification or categorization behaviors of humans in an organizational environment, the primary contributions will be in the understanding of categorization in context and in an improved understanding of the handling of information in organizations. There will be contributions on the practical side as well.

The following Table 5 outlines possible avenues of research relating to the nine dilemmas discussed previously.

As supported by research and practice presented in the literature reviewed above, these concepts require attention if we are to effectively address current and ongoing challenges for intelligence and security informatics. The development of this understanding will aid research and practice and contribute to knowledge, both academic and practical. At the most abstract, academic level it will lead the way toward a synthesis in the theory related to this phenomenon. At the very least, it will serve to suggest other ways of looking at the problem of information security. On the practical side, the payoff would be a better method to organize (or tag) the information to save the costs of unnecessary security measures in all types of organizations.

TABLE 5. A future research agenda.

Challenges	The research agenda
Dilemma #1	Research that would support the development of categorizations and classification schemas in the federal government that would allow for greater information sharing among agencies or that would allow all-source intelligence analysts to truly have access to all sources available.
Dilemma #2	Research that would explore the world of information security systems to enable a description of the different types that exist and to begin to understand the vocabulary of each and the variety of definitions of sensitive information.
Dilemma #3	Research that would reveal how well schemas “imagined” by software and systems designers compare with the schemas in everyday use in organizations.
Dilemma #4	Research to provide ground truth (metrics) about the relative degree to which attacks or intrusions actually result in compromise or destruction of sensitive information.
Dilemma #5	Research that focuses on the many boundaries (individual, group, organization, etc.) that sensitive information crosses and the conditions under which these crossings occur. One priority would be to focus on crossings between two organizations.
Dilemma #6	Research is needed to help us understand categorizations in context and to understand classifications in praxis. A special need is to understand to what degree these classifications are formal or informal.
Dilemma #7	Research is needed to better understand the phenomenon of information ownership and to reveal how and under what conditions ownership can be expanded.
Dilemma #8	Research is needed to indicate how sensitivity judgments vary among humans and the conditions under which these judgments change over time.
Dilemma #9	Research is needed concerning basic judgments about sensitivity and how these judgments are operationalized in formal and informal classification schemas.

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